

Project Executable Files

SmartBridge | Skill Wallet Project Submission Documentation

Date	15 March 2026
Team ID	SWTID-2026-5650
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	3 Marks

This section requires submission of all executable and deployable files for your project. Ensure all files are properly organized, named, and uploaded to the specified submission platform. The evaluator must be able to run or deploy your solution using the provided files and instructions.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	Yes
5	Environment configuration file (.env.example or similar)	Yes
6	Deployed application URL (if hosted)	Yes
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA
8	Dockerfile / Containerization config (if applicable)	Yes
9	Test files and test results	Yes
10	Demo video or walkthrough (if required)	Yes

Step 2: File / Folder Structure

Overview of the project's repository and directory layout:

```
comiccraft-ai/  
├── src/  
│   ├── app.py           # Main Streamlit web application interface  
│   ├── story_generator.py # Gemini 1.5 Pro story & script generation engine  
│   ├── image_generator.py # Gemini Imagen comic panel art generator  
│   ├── layout_engine.py  # Comic grid layout & panel rendering  
│   └── utils.py          # Prompt engineering and API helper routines
```

```
■■■ config/
■   ■■■ settings.py           # Global application parameters
■■■ tests/
■   ■■■ test_story.py        # Unit tests for story script generation
■   ■■■ test_image.py        # Unit tests for panel art generation
■■■ docs/
■   ■■■ Sample Project Documentation.pdf
■   ■■■ Project Executable Files.pdf
■■■ .env.example              # Template environment configuration file
■■■ Dockerfile                # Container deployment configuration
■■■ README.md                 # Detailed setup, execution, and user guide
■■■ requirements.txt           # Python package dependencies
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	https://comiccraft-ai.streamlit.app
Login Credentials (Demo)	Public Access / Demo API Key configured
Platform / Hosting Provider	Streamlit Community Cloud / Render / AWS
Repository Link	https://github.com/SWTID-2026-5650/ComicCraft-AI-Comic-Story-Creator
Demo Video Link	https://youtube.com/watch?v=comiccraft_demo_2026

Step 4: Run Instructions

Pre-requisites: • Python 3.10 or higher installed • Google Gemini API Key (obtained from Google AI Studio) • Git version control tool installed

Step 1: Clone the repository: `git clone https://github.com/SWTID-2026-5650/ComicCraft-AI-Comic-Story-Creator.git`

Step 2: Install dependencies: `pip install -r requirements.txt`

Step 3: Configure environment variables: Create a `.env` file from `.env.example` and insert your key:
`GEMINI_API_KEY=your_google_gemini_api_key_here`

Step 4: Launch application: `streamlit run src/app.py`

Step 5: Open browser at `http://localhost:8501` to access ComicCraft.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Gemini API rate limiting on free tier during heavy requests	Implemented script caching and exponential backoff retry mechanism.
2	Minor visual character inconsistency across complex panel prompts	Injected detailed character visual descriptor anchors into every prompt.
3	Image rendering latency for multi-panel comic spreads	Added asynchronous panel generation with real-time UI progress bars.